Medical Insight Explorer Agent

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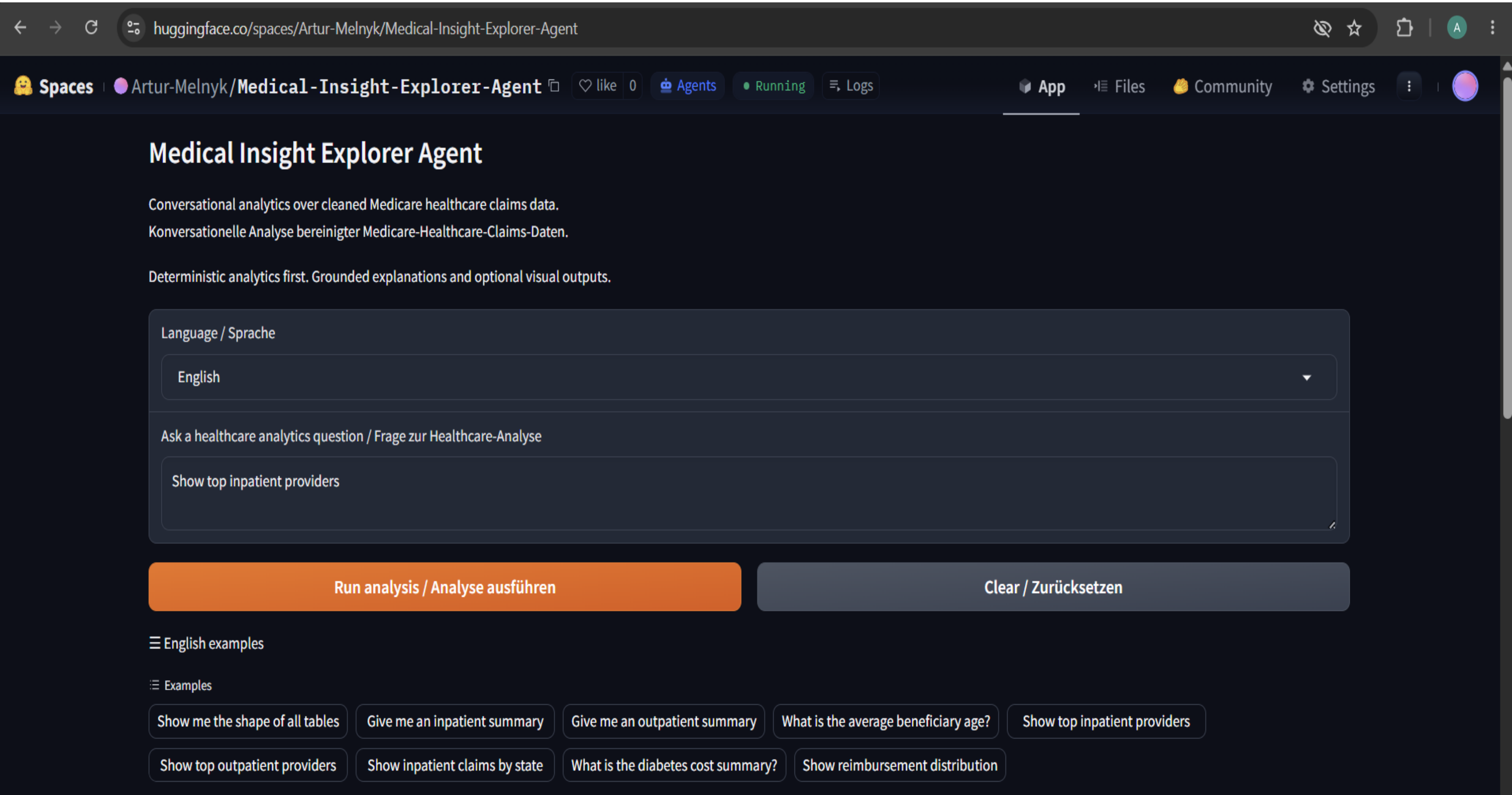
Bilingual conversational healthcare analytics with deterministic insights.

LangGraph + Gradio + Hugging Face



Business Problem

- Healthcare claims data is large and difficult to explore
- Analysts need grounded and deterministic outputs
- LLM systems in healthcare require explainability and safety
- Goal: fast analytics without clinical advice

Project Ecosystem

Healthcare Data Cleaning Repo
↓
Validated Parquet Data
↓
Medical Insight Explorer Agent
↓
Gradio + LangGraph
↓
Hugging Face Deployment

Architecture

Healthcare Data Cleaning Repository
↓
Validated Parquet Tables
↓
HealthcareDataLoader
↓
HealthcareAnalyticsEngine
↓
VisualizationTools
↓
ResponseGenerator
↓
LangGraph Workflow
↓
Bilingual Gradio Interface
↓
Hugging Face Deployment

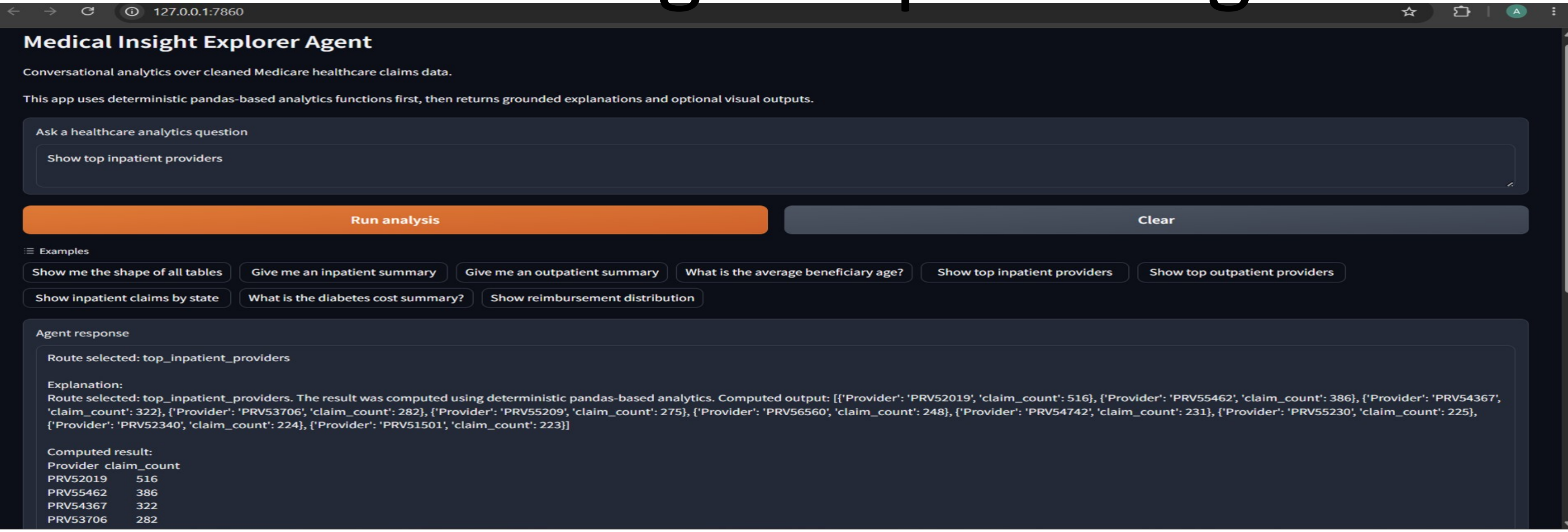
Technical Design Choices

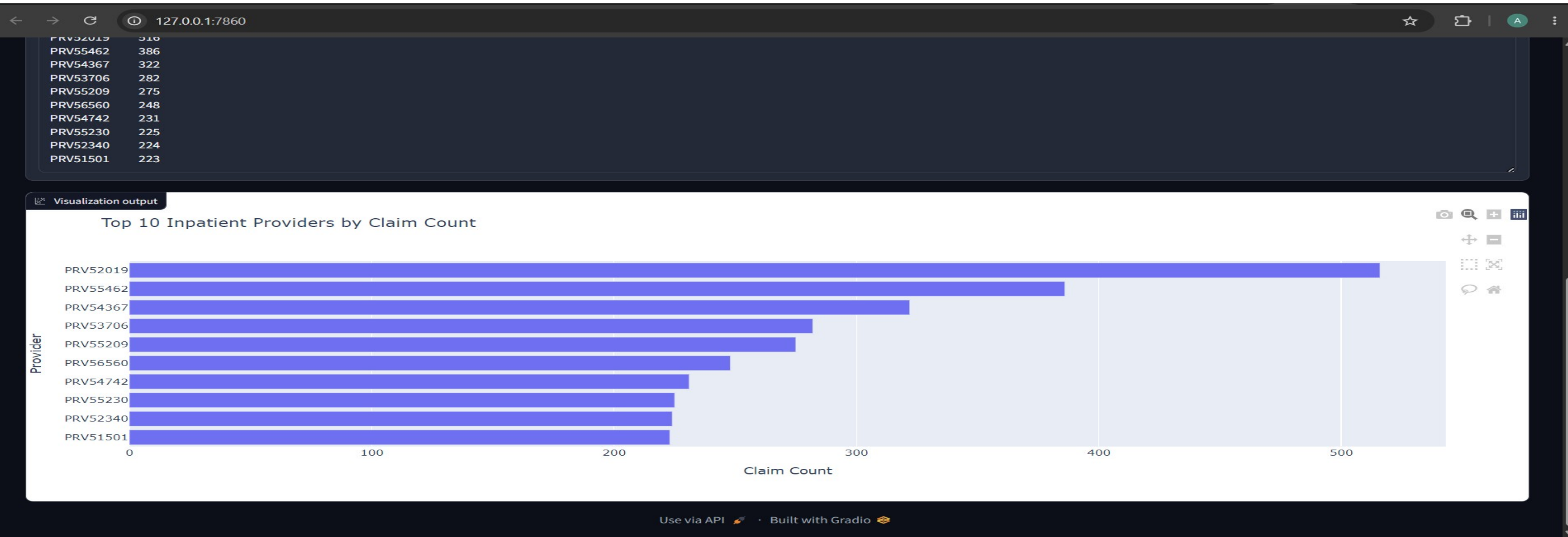
Engineering decisions used to improve reliability, safety, and deployment stability

Design Decision	Why it was implemented
Compute First → Explain Second	Deterministic pandas analytics run before AI explanations to prevent fabricated healthcare statistics
Controlled Analytics Routing	User questions are routed only to predefined functions; the LLM never receives direct DataFrame access
Sample Data Deployment Strategy	Lightweight sample datasets were used to satisfy Hugging Face memory and runtime constraints
Localized Bilingual UX	English and German interaction improves accessibility and recruiter usability
Conditional Visualization Routing	Plotly charts are generated only for supported analytical routes to avoid misleading outputs

Design principle: Safe computation → controlled reasoning → explainable output

Medical Insight Explorer Agent

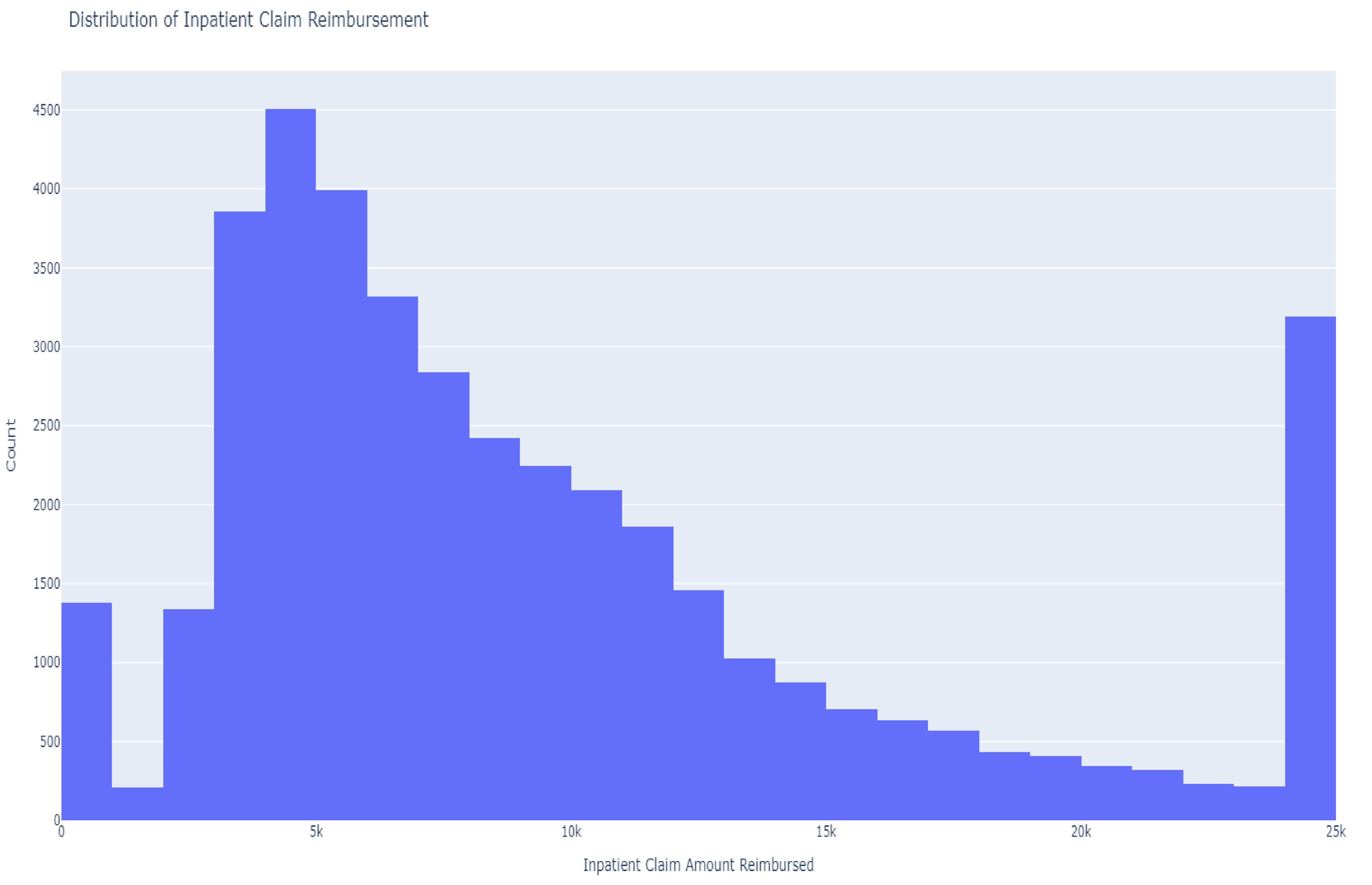




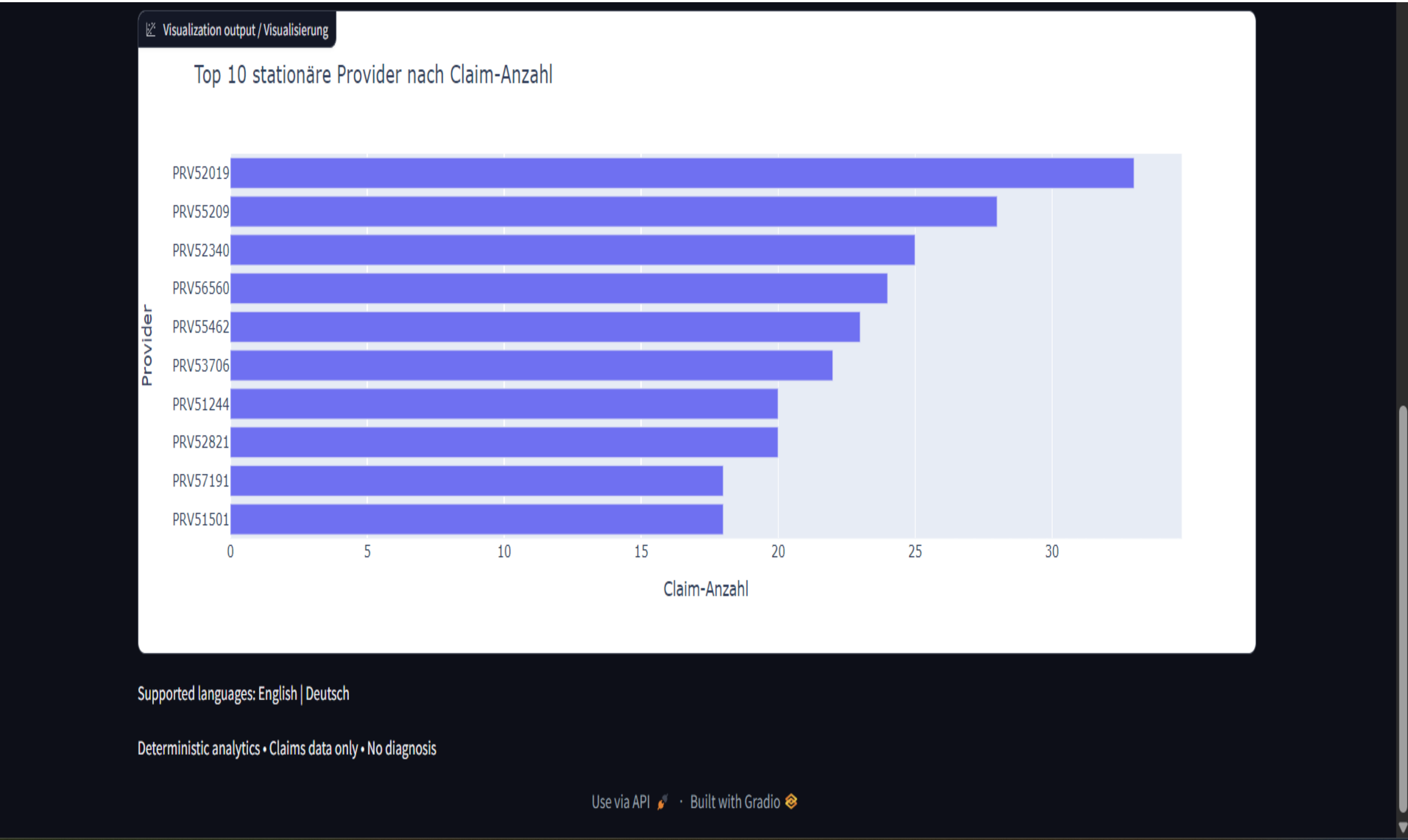
Key Features

✓ English/German UX
✓ Plotly charts
✓ Deterministic analytics
✓ LangGraph orchestration
✓ Safe compute-first architecture

English Demo

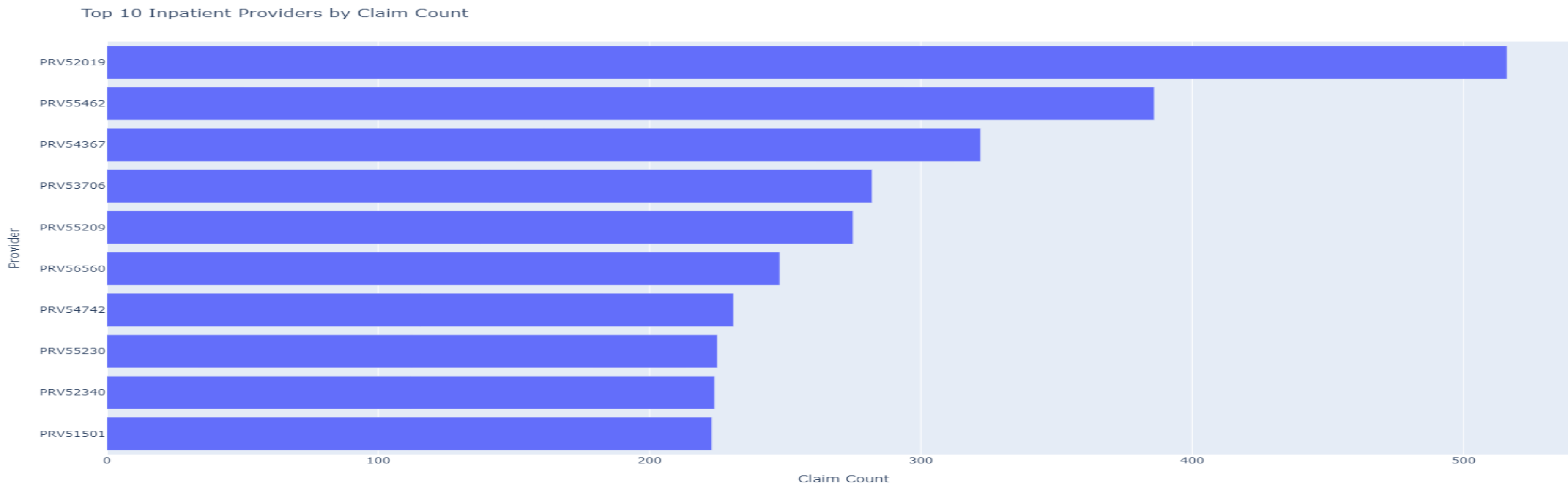


German Demo

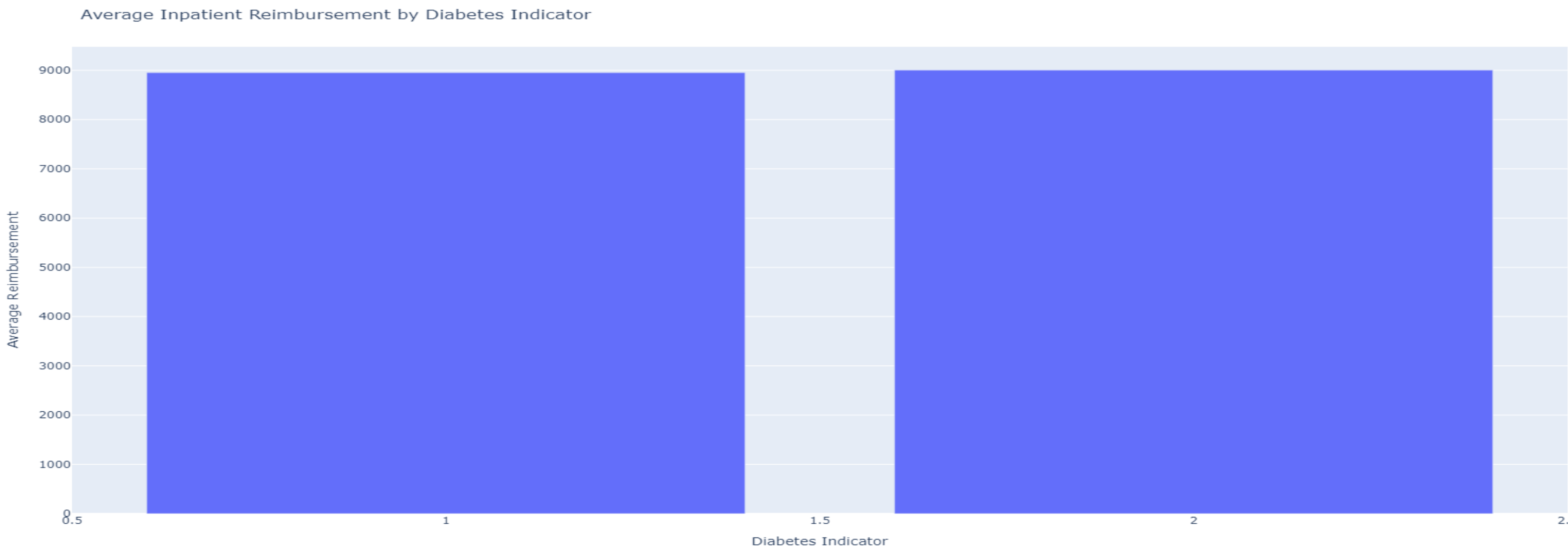


Visualization output / Visualisierung
Top 10 stationäre Provider nach Claim-Anzahl
Provider
PRV52019
PRV55209
PRV52340
PRV56560
PRV55462
PRV53706
PRV51244
PRV52821
PRV57191
PRV51501
0
5
10
15
20
25
30
Claim-Anzahl
Supported languages: English | Deutsch
Deterministic analytics • Claims data only • No diagnosis
Use via API · Built with Gradio

Analytics Outputs



Insight: Provider activity is concentrated among a small subset of institutions.



Insight: Reimbursements are right-skewed with a concentration around lower claim amounts.

Safety & Responsible AI

✓ Claims analytics only
✓ No diagnosis
✓ No treatment recommendations
✓ Explainable deterministic calculations
✓ Safer healthcare AI interaction

Live Demo & Portfolio

Key Takeaways

✓ Safe healthcare analytics
✓ Explainable AI workflow
✓ Production deployment
✓ Bilingual recruiter-ready UX

Hugging Face:
https://huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent

GitHub:
https://github.com/ArturMelnyk-analyst/Medical-Insight-Explorer-Agent

Built using: Python • Pandas • Plotly • Gradio • LangGraph • Hugging Face